

NETWORK TRAFFIC ANALYSIS USING WIRESHARK

Internship Project Report

Submitted By: Adon Johnson

Course : IMCA

Rajagiri College of Social Sciences, Kalamassery

Submitted To:

CodeC Technologies

INTRODUCTION

Network Traffic Analysis is the process of observing and understanding the data that travels through a network. Whenever we open a website, watch a YouTube video, or search on Google, data packets are sent and received through the internet. In this project, I used Wireshark software to capture and analyze network traffic. This helped me understand how different protocols work and how devices communicate over a network.

OBJECTIVE

- To understand the basics of network traffic analysis.
- To learn packet capturing using Wireshark.
- To analyze DNS, TCP and TLS protocols.
- To gain practical experience in network monitoring.

TOOLS USED

Wireshark is a packet analyzer used for capturing and analyzing network traffic.

tcpdump is a command-line tool used to capture network packets.

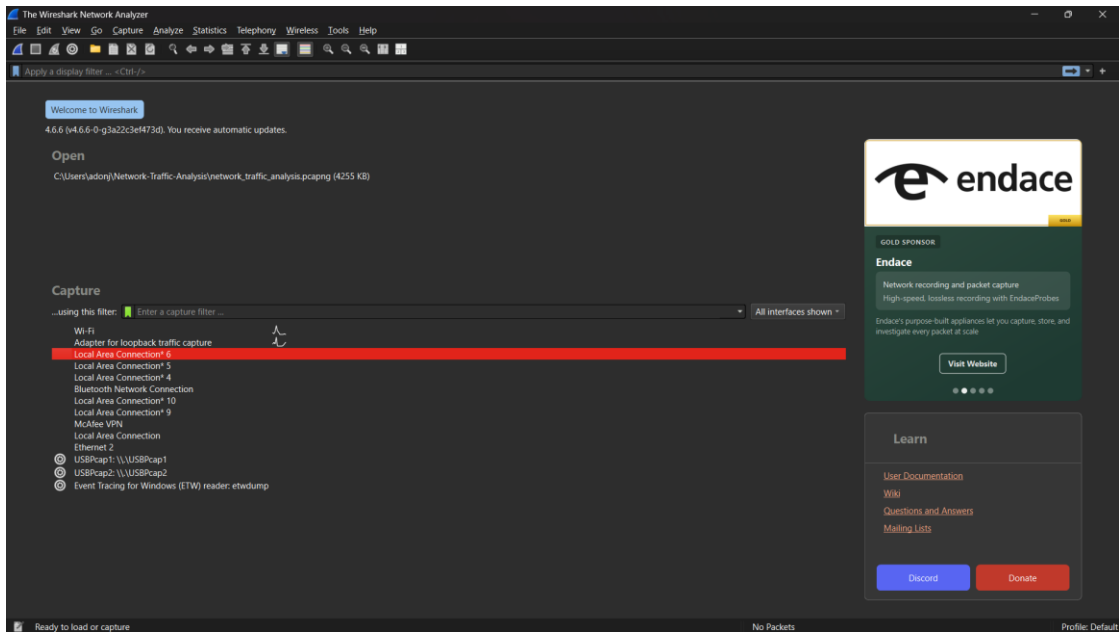
Snort is an intrusion detection tool used for identifying suspicious network activities.

For this project, Wireshark was mainly used.

PROCEDURE

1. Installed Wireshark and opened the application.
2. Selected the Wi-Fi interface and started packet capture.
3. Generated traffic by searching on Google, opening YouTube and visiting websites.
4. Stopped the capture after a few minutes.
5. Applied DNS, TCP and TLS filters for analysis.

Screenshot 1 – Wireshark Home Screen



This home screen shows the available network interfaces used for packet capture.

Screenshot 2 – Captured Network Traffic

The screenshot displays the Wireshark interface with a network traffic capture. The main pane shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, and Length. The packet list includes various TCP connections, including keep-alive messages and application data. A packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (query). The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
122162	428.769235700	2409:40f3:20:97a3:1...	64:ff9b::36f:f1e0	TCP	75	[TCP Keep-Alive] 64375 → 443 [ACK] Seq=5033 Ack=8068 Win=65280 Len=1 [TCP PDU reassembled in 66059]
122163	428.812275200	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 60733 [ACK] Seq=7598 Ack=3501 Win=69632 Len=0 SLE=3500 SRE=3501
122164	428.812275600	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 54659 [ACK] Seq=7599 Ack=3469 Win=69632 Len=0 SLE=3468 SRE=3469
122165	428.812275900	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 53998 [ACK] Seq=9006 Ack=8232 Win=110592 Len=0 SLE=8231 SRE=8232
122166	428.812276200	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 57742 [ACK] Seq=7599 Ack=3533 Win=53248 Len=0 SLE=3532 SRE=3533
122167	428.818176800	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 64375 [ACK] Seq=8068 Ack=5034 Win=90112 Len=0 SLE=5033 SRE=5034
122168	429.059826500	64:ff9b::1487:4a9	2409:40f3:20:97a3:1...	TCP	74	443 → 60236 [RST, ACK] Seq=12441 Ack=3392 Win=0 Len=0
122169	429.183866200	2409:40f3:20:97a3:1...	2405:200:1607:1731:...	QUIC	91	Protected Payload (KPB), DCID=03adf14e7b00811c
122170	429.162618100	2405:200:1607:1731:...	2409:40f3:20:97a3:1...	QUIC	86	Protected Payload (KPB)
122171	429.161535400	10.155.162.207	172.16.0.1	TCP	60	443 → 55139 [SYN] Seq=0 Win=0 MSS=1460 WS=256 SACK_PERM
122172	429.665975400	2409:40f3:20:97a3:1...	2405:200:1630:1400:...	TCP	75	[TCP Keep-Alive] 53644 → 443 [ACK] Seq=3193 Ack=210428 Win=130560 Len=1
122173	429.699739800	2405:200:1630:1400:...	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 53644 [ACK] Seq=210428 Ack=3194 Win=64128 Len=0 SLE=3193 SRE=3194
122174	429.713956100	2409:40f3:20:97a3:1...	2406:4700:83b3:9532	TCP	75	[TCP Keep-Alive] 54449 → 443 [ACK] Seq=2426 Ack=11014 Win=64256 Len=1
122175	429.751801400	2406:4700:83b3:9532	2409:40f3:20:97a3:1...	TCP	86	[TCP Keep-Alive ACK] 443 → 54449 [ACK] Seq=11014 Ack=2427 Win=131872 Len=0 SLE=2426 SRE=2427
122176	430.343315700	2603:1040:a06:3:13	2409:40f3:20:97a3:1...	TLSv1.2	101	Application Data
122177	430.387322700	2409:40f3:20:97a3:1...	2603:1040:a06:3:13	TCP	74	63415 → 443 [ACK] Seq=1261 Ack=757 Win=254 Len=0
122178	430.771739800	2409:40f3:20:97a3:1...	2405:200:1607:1731:...	QUIC	91	Protected Payload (KPB), DCID=03adf14e7b00811c
122179	430.806738700	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	74	[TCP Keep-Alive] 443 → 64375 [ACK] Seq=8067 Ack=5034 Win=90112 Len=0
122180	430.806739500	64:ff9b::36f:f1e0	2409:40f3:20:97a3:1...	TCP	74	[TCP Keep-Alive] 443 → 53998 [ACK] Seq=9005 Ack=8232 Win=110592 Len=0
122181	430.806786400	2409:40f3:20:97a3:1...	64:ff9b::36f:f1e0	TCP	74	[TCP Keep-Alive ACK] 64375 → 443 [ACK] Seq=5034 Ack=8068 Win=65280 Len=0

Frame 1: Packet, 85 bytes on wire (680 bits), 85 bytes captured (680 bits) on interface \Device\NPF...
Ethernet II, Src: AzureWavelec, 36:c8:97:14:33:33, 16:c8:97, Dst: ba:b6:00:af:70:30 (ba:b6:00:af:70:30)
Internet Protocol Version 4, Src: 10.155.162.207, Dst: 10.155.162.201
User Datagram Protocol, Src Port: 55139, Dst Port: 53
Domain Name System (query)

This screenshot shows live packets captured during browsing activity.

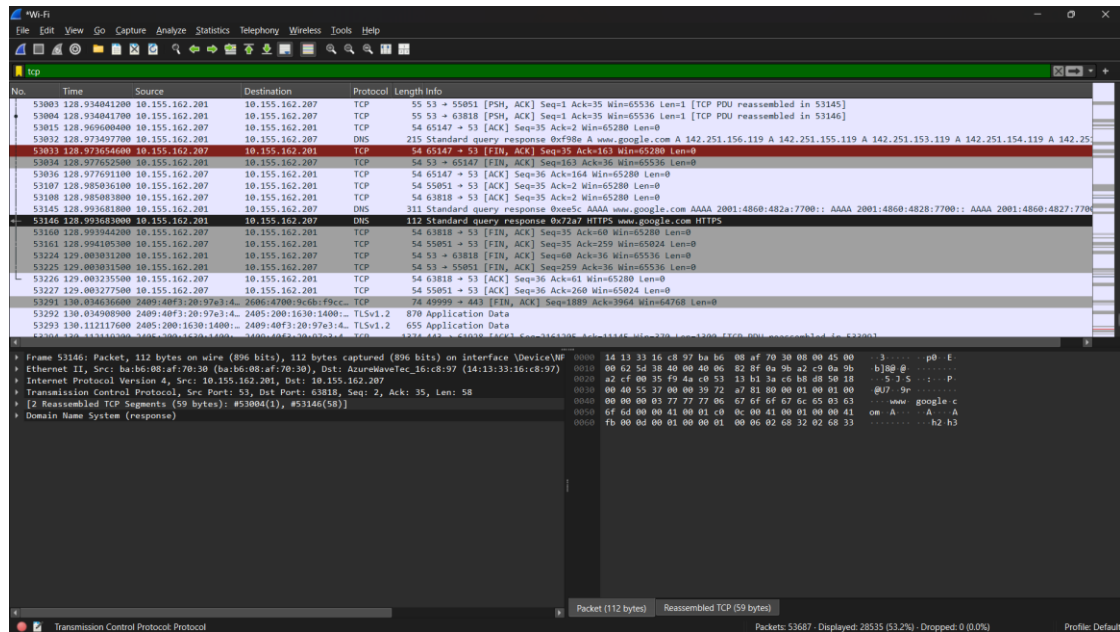
Screenshot 3 – DNS Analysis

The screenshot displays a Wireshark network traffic capture with the DNS filter applied. The packet list shows a series of DNS queries and responses from source IP 10.155.162.201 to destination IP 10.155.162.207. The queries are for various domains including lh3.googleusercontent.com, drive.google.com, fonts.gstatic.com, and www.google.com. The responses provide IP addresses for these domains. The packet details pane shows the structure of a DNS Standard Query Response, including the header, question section, answer section, and authority section. The packet bytes pane shows the raw data of the captured packet.

No.	Time	Source	Destination	Protocol	Length	Info
52003	123.261631300	10.155.162.201	10.155.162.207	DNS	184	Standard query response 0xeadd HTTPS lh3.googleusercontent.com CNAME googlehosted.l.googleusercontent.com SOA ns1.google.com
52004	123.261631800	10.155.162.201	10.155.162.207	DNS	143	Standard query response 0x6a69 A lh3.googleusercontent.com CNAME googlehosted.l.googleusercontent.com A 142.251.222.193
52245	125.845668000	10.155.162.201	10.155.162.207	DNS	88	Standard query 0xa20f A drive.google.com
52251	125.845854400	10.155.162.201	10.155.162.207	DNS	88	Standard query 0x0018 AAAA drive.google.com
52253	125.845927600	10.155.162.201	10.155.162.207	DNS	88	Standard query 0x6c2b HTTPS drive.google.com
52274	125.162898500	10.155.162.201	10.155.162.207	DNS	139	Standard query response 0x6c2b HTTPS drive.google.com SOA ns1.google.com
52275	125.163066000	10.155.162.201	10.155.162.207	DNS	105	Standard query response 0xa20f A drive.google.com A 142.251.223.174
52276	125.163070600	10.155.162.201	10.155.162.207	DNS	117	Standard query response 0x0018 AAAA drive.google.com AAAA 2404:6800:4007:821::200e
52778	126.865382300	10.155.162.201	10.155.162.207	DNS	89	Standard query 0x4059 A fonts.gstatic.com
52780	126.865418900	10.155.162.201	10.155.162.207	DNS	89	Standard query 0x0308 AAAA fonts.gstatic.com
52782	126.865455700	10.155.162.201	10.155.162.207	DNS	89	Standard query 0x0a50 HTTPS fonts.gstatic.com
52795	126.990831600	10.155.162.201	10.155.162.207	DNS	118	Standard query response 0x0308 AAAA fonts.gstatic.com AAAA 2404:6800:4007:815::2003
52796	126.990832100	10.155.162.201	10.155.162.207	DNS	115	Standard query response 0x0a50 HTTPS fonts.gstatic.com HTTPS
52797	126.990832600	10.155.162.201	10.155.162.207	DNS	106	Standard query response 0x4059 A fonts.gstatic.com A 142.251.43.35
52977	128.854074500	10.155.162.201	10.155.162.207	DNS	86	Standard query 0x72a7 HTTPS www.google.com
52979	128.854146400	10.155.162.201	10.155.162.207	DNS	86	Standard query 0xf98e A www.google.com
52981	128.854199700	10.155.162.201	10.155.162.207	DNS	86	Standard query 0xe5c AAAA www.google.com
53032	128.973497700	10.155.162.201	10.155.162.207	DNS	215	Standard query response 0xf98e A www.google.com A 142.251.156.119 A 142.251.155.119 A 142.251.153.119 A 142.251.154.119 A 142.251.155.119
53145	128.993681800	10.155.162.201	10.155.162.207	DNS	311	Standard query response 0xe5c AAAA www.google.com AAAA 2001:4860:482a:7700::AAAA 2001:4860:4828:7700::AAAA 2001:4860:4827:7700::AAAA
53146	128.993683000	10.155.162.201	10.155.162.207	DNS	112	Standard query response 0x72a7 HTTPS www.google.com HTTPS

DNS converts domain names into IP addresses. DNS requests and responses were successfully captured.

Screenshot 4 – TCP Analysis



The TCP filter was applied to view TCP packets. Different TCP packets such as ACK and FIN packets were observed, showing communication between the local system and remote servers.

Screenshot 5 – TLS Analysis

The screenshot displays a Wireshark packet capture analysis of network traffic. The packet list pane shows a series of packets, with the selected packet (No. 52965) being a TLS packet. The packet details pane shows the structure of the selected packet, including the Frame, Ethernet II, Internet Protocol Version 6, Transmission Control Protocol, and Transport Layer Security (TLS) sections. The packet bytes pane shows the raw hex and ASCII data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
52325	125.326279900	2404:6800:4007:821::	2409:40f3:20:97a3:4::	TLSv1.3	1294	Change Cipher Spec
52328	125.326294400	2404:6800:4007:821::	2409:40f3:20:97a3:4::	TLSv1.3	297	Application Data
52330	125.327186700	2409:40f3:20:97a3:4::	2404:6800:4007:821::	TLSv1.3	148	Change Cipher Spec, Application Data
52331	125.363684700	2404:6800:4007:821::	2409:40f3:20:97a3:4::	TLSv1.3	1036	Application Data, Application Data
52803	126.993602700	2409:40f3:20:97a3:4::	2404:6800:4007:815::	QUIC	1292	Initial, OCID=c1935222b78cd816, PKN: 3, PADDING, CRYPTO, PING, PING, PADDING, PING, PING, CRYPTO, CRYPTO, PADDING, PING, CRYPTO,
52816	127.064752400	2404:6800:4007:815::	2409:40f3:20:97a3:4::	QUIC	1292	Initial, SCID=c1935222b78cd816, PKN: 5, CRYPTO, PADDING
52926	128.251236900	2600:1040:a06:3:1:13	2409:40f3:20:97a3:4::	TLSv1.2	101	Application Data
52927	128.266788700	2606:4700:9c61:d9b4::	2409:40f3:20:97a3:4::	TLSv1.2	98	Application Data
52928	128.266985500	2409:40f3:20:97a3:4::	2606:4700:9c61:d9b4::	TLSv1.2	102	Application Data
52953	128.408954700	2409:40f3:20:97a3:4::	2606:1901:1:d18::	TLSv1.2	117	Application Data
52965	128.668378200	2606:1901:1:d18::	2409:40f3:20:97a3:4::	TLSv1.2	114	Application Data
53292	130.034908900	2409:40f3:20:97a3:4::	2405:200:1630:1400::	TLSv1.2	870	Application Data
53293	130.112117000	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	655	Application Data
53300	130.127704000	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	880	Application Data
53328	130.138844100	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	1374	Application Data
53334	130.141286500	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	162	Application Data
53353	130.142426400	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	1374	Application Data
53366	130.142432700	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	1374	Application Data
53372	130.142437000	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	1374	Application Data
53386	130.143126900	2405:200:1630:1400::	2409:40f3:20:97a3:4::	TLSv1.2	1374	Application Data

Packet 52965 details:

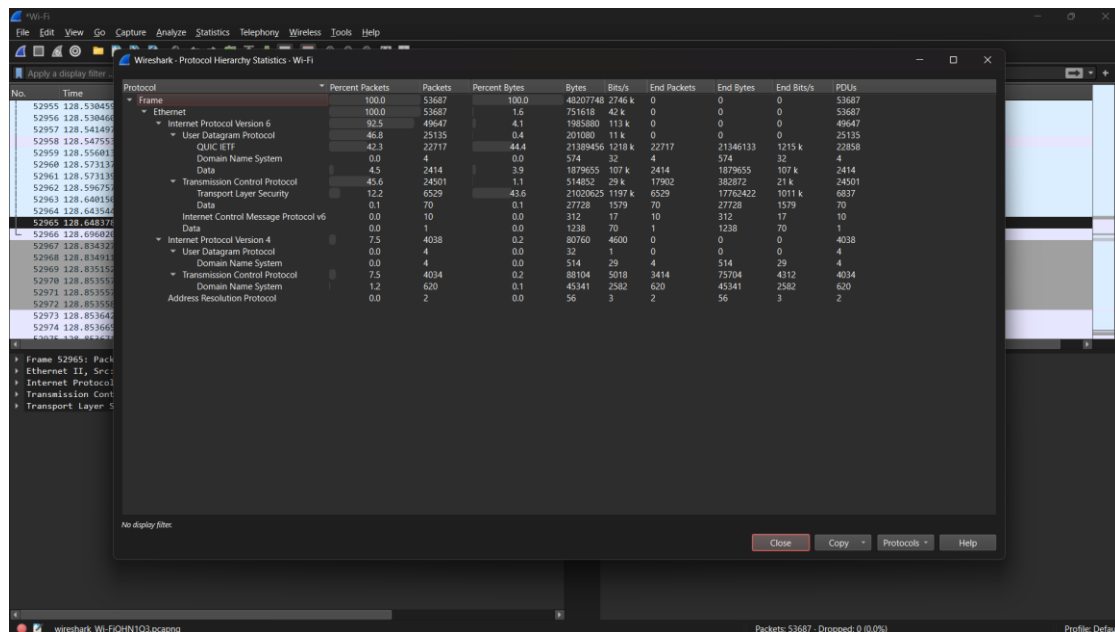
- Frame 52965: Packet, 114 bytes on wire (912 bits), 114 bytes captured (912 bits) on interface DeviceVM
- Ethernet II, Src: ba:16:00:af:70:30 (ba:16:00:af:70:30), Dst: AzureWaveTec_16:ca:97 (14:13:33:16:ca:97)
- Internet Protocol Version 6, Src: 2606:1901:1:d18::, Dst: 2409:40f3:20:97a3:4::, Seq: 443, Port: 52334, Seq: 161, Ack: 216, Len: 40
- Transmission Control Protocol, Src Port: 443, Dst Port: 52334, Seq: 161, Ack: 216, Len: 40
- Transport Layer Security

Packet bytes:

```
0000 14 13 33 16 ca 97 ba b6 00 af 70 30 86 dd 66 8d 3  . . . p b f
0010 7f 52 00 3c 86 78 26 00 19 01 00 01 0d 18 00 00 8 c x
0020 00 00 00 00 00 24 09 40 f3 00 20 97 e3 45 aa . . . $ . . E
0030 fd c0 e8 92 e8 5d 01 bb cc 6e 8d 44 86 b7 85 fb . . . ] . n D .
0040 77 9b 50 18 04 09 e3 27 00 00 17 03 00 23 ce w P . . . #
0050 d6 ae 17 d0 ea 2b 72 55 b6 e4 02 08 19 55 b2 79 + r U . . . U y
0060 2a e9 7b 8f de f5 5a 19 8d 54 d9 98 6f a6 ec b9 * { . Z . T . o .
0070 5a df z
```

During the packet capture process, different types of network traffic were observed. DNS packets were used for domain name resolution. TCP packets were responsible for reliable communication between devices. TLS packets showed encrypted communication with websites.

Screenshot 6 – Protocol Hierarchy Statistics



The protocol hierarchy displays IPv4, IPv6, TCP, UDP, DNS, QUIC and TLS traffic statistics.

ANALYSIS

DNS packets related to Google services were captured. TCP packets showed connection establishment and data transfer. TLS packets confirmed that communication was encrypted. The protocol hierarchy statistics showed that most traffic used IPv6, TCP, QUIC and TLS protocols.

RESULT

The project was successfully completed using Wireshark. Network packets were captured and analyzed using different filters. DNS, TCP, TLS, IPv4 and IPv6 protocols were identified. The captured traffic helped in understanding real-world network communication and security mechanisms..

CONCLUSION

This project helped me understand how data travels through a network and how different protocols are used during communication. By using Wireshark, I gained practical experience in packet capturing and analysis. The project improved my understanding of networking concepts and basic cybersecurity practices.

REFERENCES

1. Wireshark Official Documentation
2. Internship Learning Materials
3. Basic Networking Notes
4. Online Learning Resources